

GenAI Final Project

Tool-16 — SOC 2 Readiness Manager

Role-AI Developer 2

Introduction

This project focuses on developing an AI-powered SOC2 Readiness Manager system. It leverages the Groq API to process user inputs and generate intelligent outputs. The system is designed to classify issues into relevant SOC2 compliance categories. It also provides detailed compliance reports including risks and recommendations.

This project presents an AI-powered SOC2 Readiness Manager designed to assist in identifying and analyzing compliance-related issues. The system leverages artificial intelligence to classify problems into appropriate SOC2 categories and generate structured compliance reports.

The application is built using Flask as the backend framework, which handles API requests and manages the overall workflow. The Groq API is integrated to process user inputs and generate intelligent responses based on prompt engineering techniques.

The system provides key functionalities such as issue categorization, report generation, and service health monitoring. These features help automate the process of compliance analysis, reducing manual effort and improving efficiency.

Additionally, the project focuses on designing clear and structured prompts to ensure accurate AI outputs. Testing and validation were performed using multiple input cases to verify the correctness and reliability of the system.

Overall, this project demonstrates the practical implementation of AI in compliance management using a simple and efficient architecture.

- The backend is implemented using Flask to handle API requests efficiently.
- This project is an AI-powered SOC2 Readiness Manager.
- It uses Groq API to generate intelligent responses.
- The system classifies issues into SOC2 categories.
- It also generates compliance reports based on input data.
- Flask is used to build and run the backend APIs.

Role:AI Developer 2

Week 1-AI Setup & Integration

- Generated and configured Groq API key using .env file
- Integrated Groq API into the backend for AI-based responses
- Developed Flask application (app.py) to handle API requests
- Created /categorise endpoint to classify issues into SOC2 categories
- Tested the API with sample inputs to verify correct AI output

Week 2- AI Feature Development

- Improved prompt design to get more accurate and structured AI responses
- Developed /generate-report endpoint to create SOC2 compliance reports
- Implemented report generation including summary, risks, and recommendations
- Added /health endpoint to check service status
- Tested APIs with multiple inputs to ensure correct functionality

Week 3- Testing & Validation

- Performed testing on all endpoints (/ , /categorise, /generate-report, /health)
- Used different input cases to validate AI performance
- Improved response clarity and formatting
- Fixed minor errors and ensured smooth execution
- Verified overall system stability and reliability

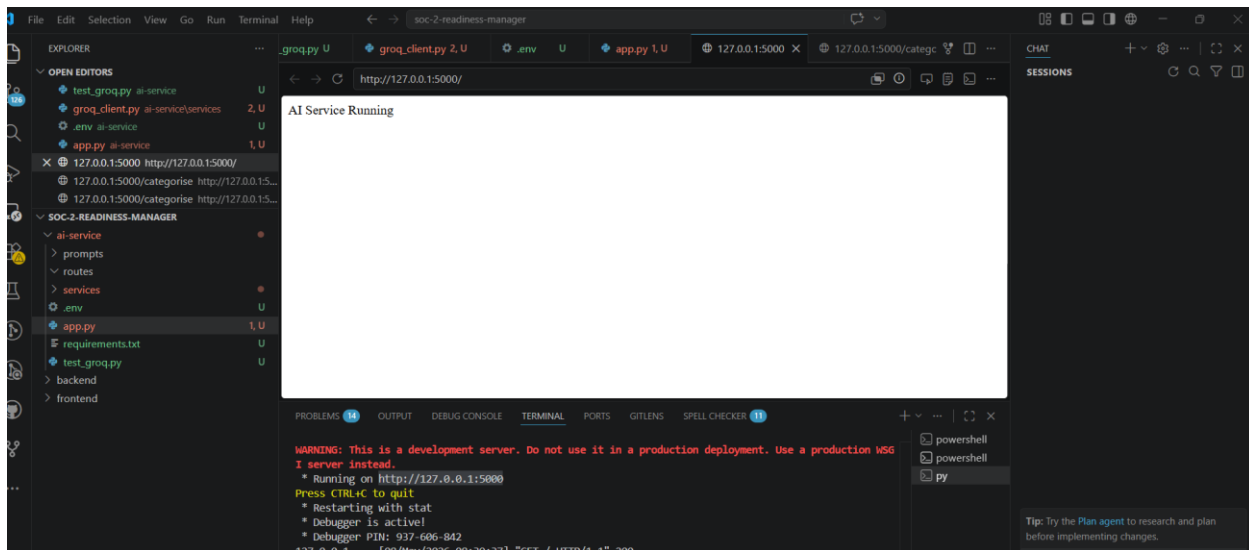
Week 4- Finalization & Demo Preparation

- Conducted final verification of all API endpoints
- Captured output screenshots for documentation
- Prepared project documentation (README and report)
- Created AI Summary Card with project details and workflow
- Practiced demo explanation of system architecture (Flask + Groq API)
- Added prompt engineering explanation in simple ter

Output Screenshots

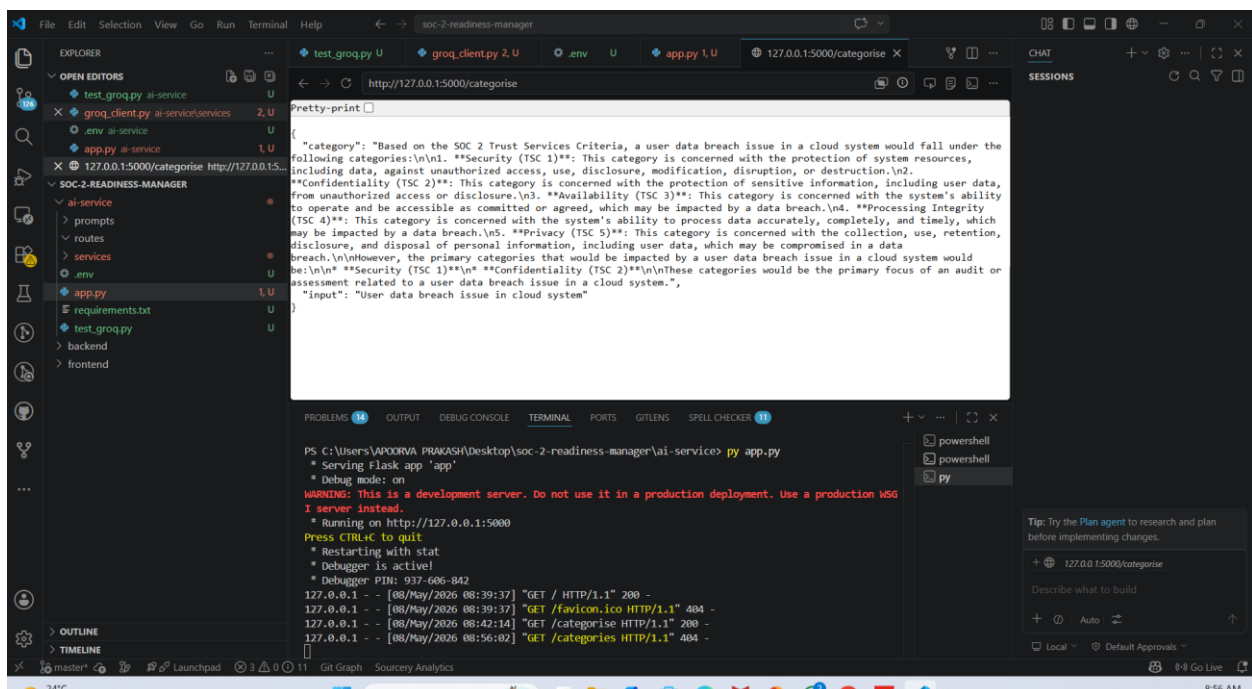
1.home.png

<http://127.0.0.1:5000/>



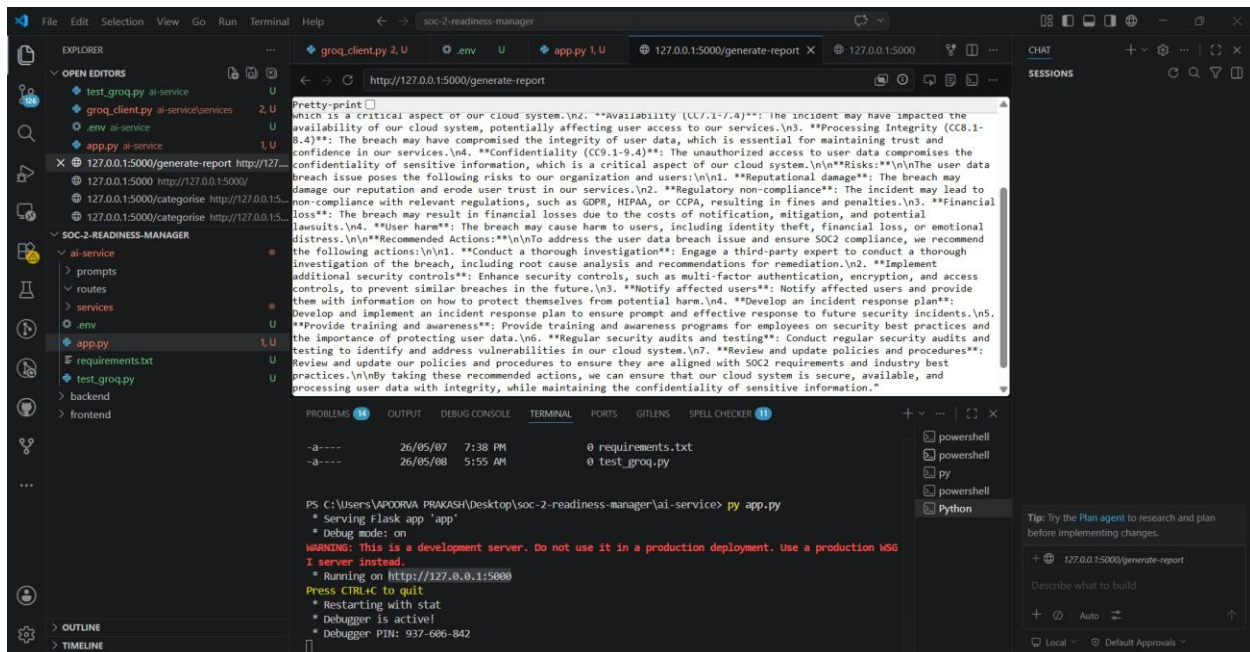
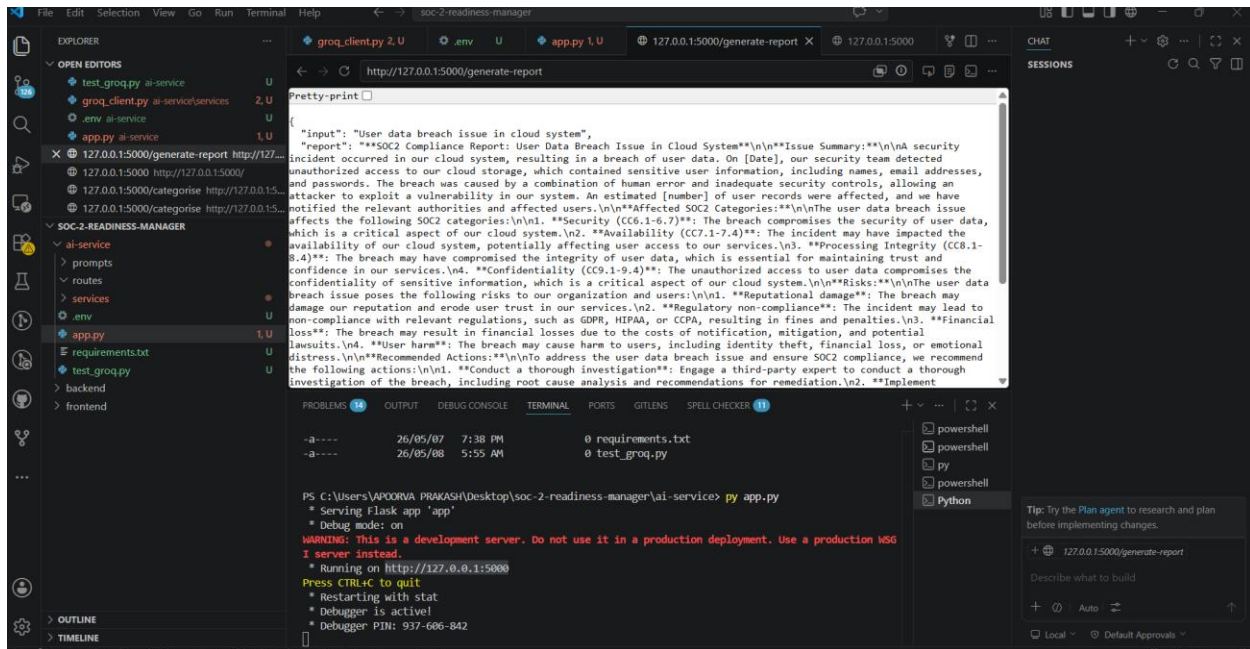
2.categorise.png

<http://127.0.0.1:5000/categorise>



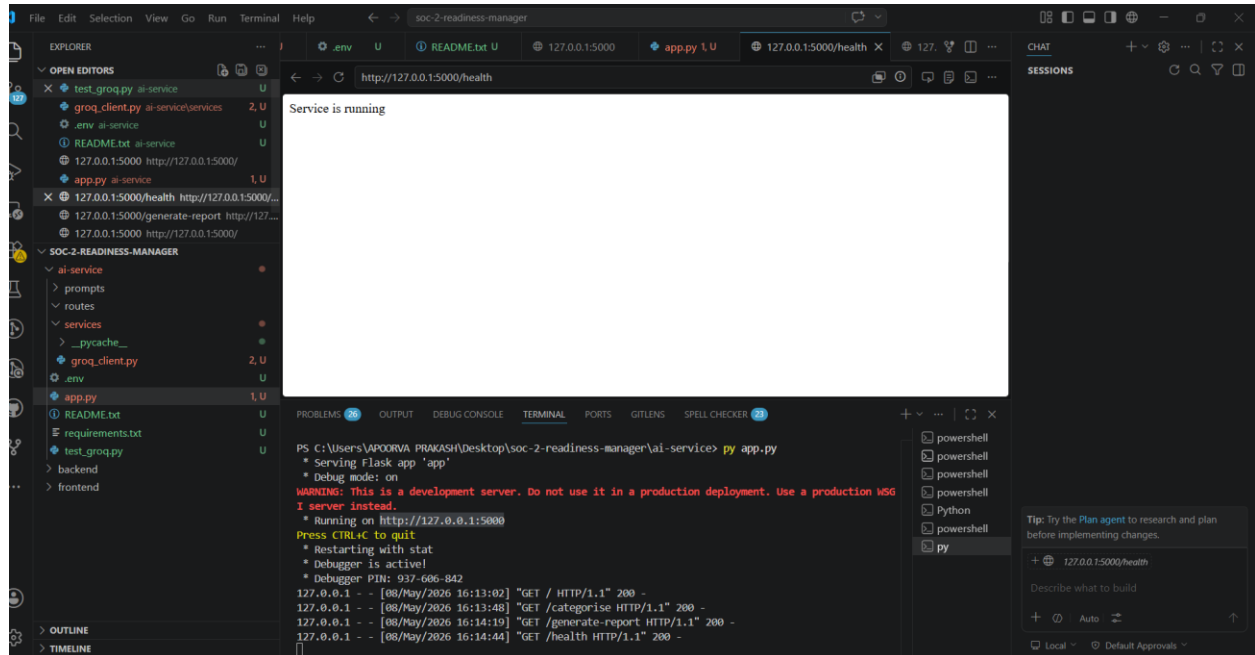
3. generate_report.png

http://127.0.0.1:5000/generate-report



4. health.png

<http://127.0.0.1:5000/health>



Prompt Engineering

Prompt engineering is the process of giving clear and structured instructions to an AI model to get accurate results.

In this project, I improved prompts by making them simple, specific, and well-formatted to generate better and more reliable outputs.

AI Summary Card

This project is an AI-powered SOC2 Readiness Manager developed using Python, Flask, and Groq API.

It is designed to automate the process of identifying and analyzing compliance-related issues.

The system takes user input and classifies it into appropriate SOC2 categories using AI.

It also generates detailed compliance reports including issue summary, risks, and recommended actions.

The application provides multiple API endpoints such as categorisation, report generation, and health monitoring to ensure smooth functionality.

Prompt engineering techniques are used to improve the quality and accuracy of AI responses.

The system follows a simple workflow: user input → AI processing → structured output.

Overall, this project demonstrates how artificial intelligence can be used to simplify compliance management and improve efficiency in real-world applications.